

University of Dhaka

Department of Computer Science & Engineering

CSE 2206: Microcontroller and Embedded Systems

ASSIGNMENT 04

ADC Multi-Resolution Testing with Flash-Based Data Logging

Semester	2nd Year, 2nd Semester 2026
Course Code & Title	CSE 2206 — Microcontroller & Embedded System
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Assignment	04
Tasks Covered	Flash Memory-Based Data Logging
Topic	Flash Memory
Platform	STM32F446RE Nucleo-64
IDE	STM32CubeIDE (Bare-Metal & HAL)
Group	B1
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1. Objective & Summary

This report documents the implementation and testing of Lab 04 . Two implementations has been produced: a **register-level (bare-metal)** version (LAB_A), which is the assessed implementation, and a **HAL-based** version (LAB_B) built for direct comparison. The **identity block is in Flash Sector 6** .

2. System Design Overview

Flash erase is **sector granular** and removes everything in a sector. **Sector 6** holds the write-once identity block. This guarantees that re-running tests can never destroy the stored identity. As Flash Memory is Non-volatile.

Sector	Base address	Size	Purpose	Validity marker
6	0x08040000	128 KB	Student identity — written once, never erased again	0xB1010001
7	0x08060000	128 KB	Test results — erased & rewritten every run	0xCAFEBAFE

On-Flash data records (**word-aligned so each field maps onto 32-bit Flash writes**):

```
typedef struct {
    uint32_t marker;                /* 0xB1010001 if valid, 0xFFFFFFFF if
blank */
    char    registration[REG_MAXLEN]; /* null-padded string */
    char    roll[ROLL_MAXLEN];       /* null-padded string */
    char    name[NAME_MAXLEN];       /* null-padded string */
} StudentInfo_t;                  /* 64 bytes total -- word-aligned */
```

```
typedef struct {
    uint32_t marker;                /* RESULTS_MARKER once a run exists */
    float    v12;                   /* averaged voltage @ 12-bit RES */
    float    v10;                   /* averaged voltage @ 10-bit RES */
    float    v8;                    /* averaged voltage @ 8-bit RES */
    float    v6;                    /* averaged voltage @ 6-bit RES */
} ResultsRecord_t;
```

Resolution	RES bits	Max code	1-LSB @3.3V	Example use case
12-bit	00	4095	0.806 mV	Precision sensing (fine light / temperature)
10-bit	01	1023	3.23 mV	General-purpose input (volume / brightness dial)
8-bit	10	255	12.94 mV	Coarse-but-fast input (threshold / level detect)
6-bit	11	63	52.38 mV	Very coarse, high-speed (3–4 position mode selector)

Boot sequence (executed on every reset):

1. Display the identity block from Sector 6 — registration, roll, and name; or “*not yet provisioned*” if the marker is blank.

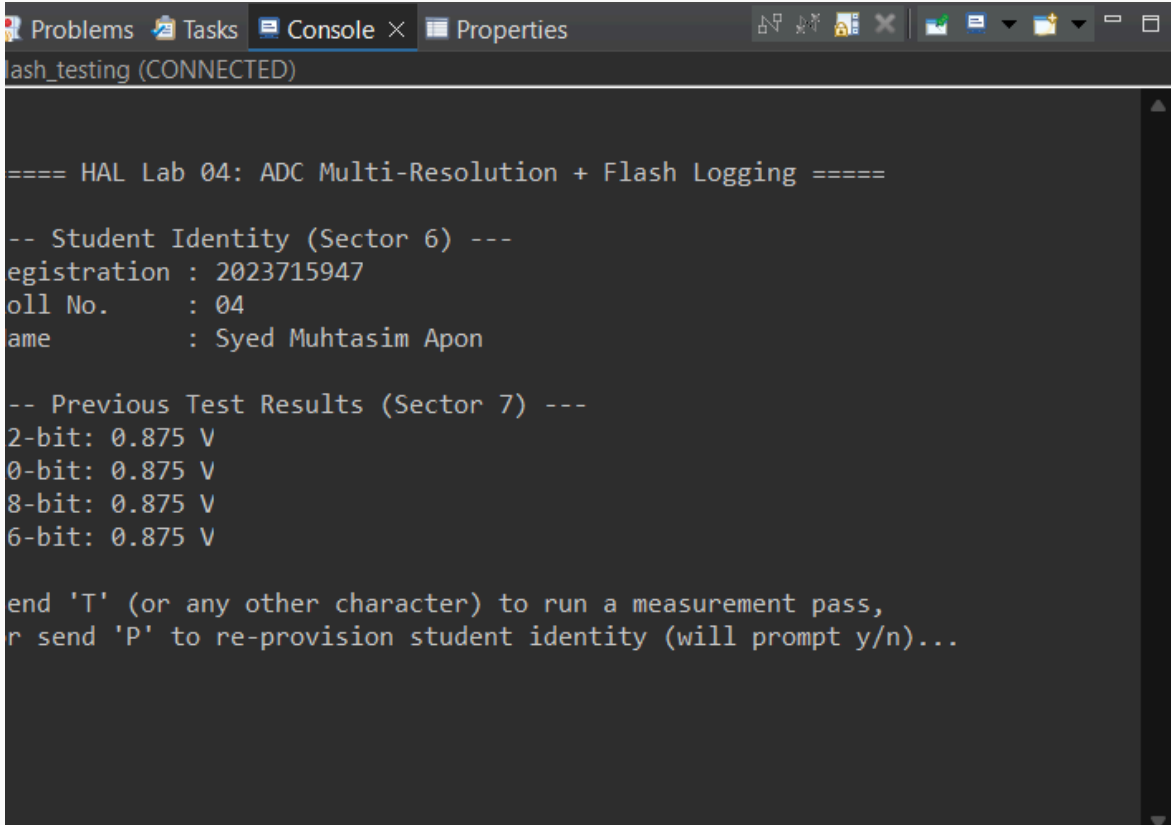
2. Display the previous test results from Sector 7, one voltage per resolution; or “no previous test data” if blank.

4. Identity-Provisioning Process

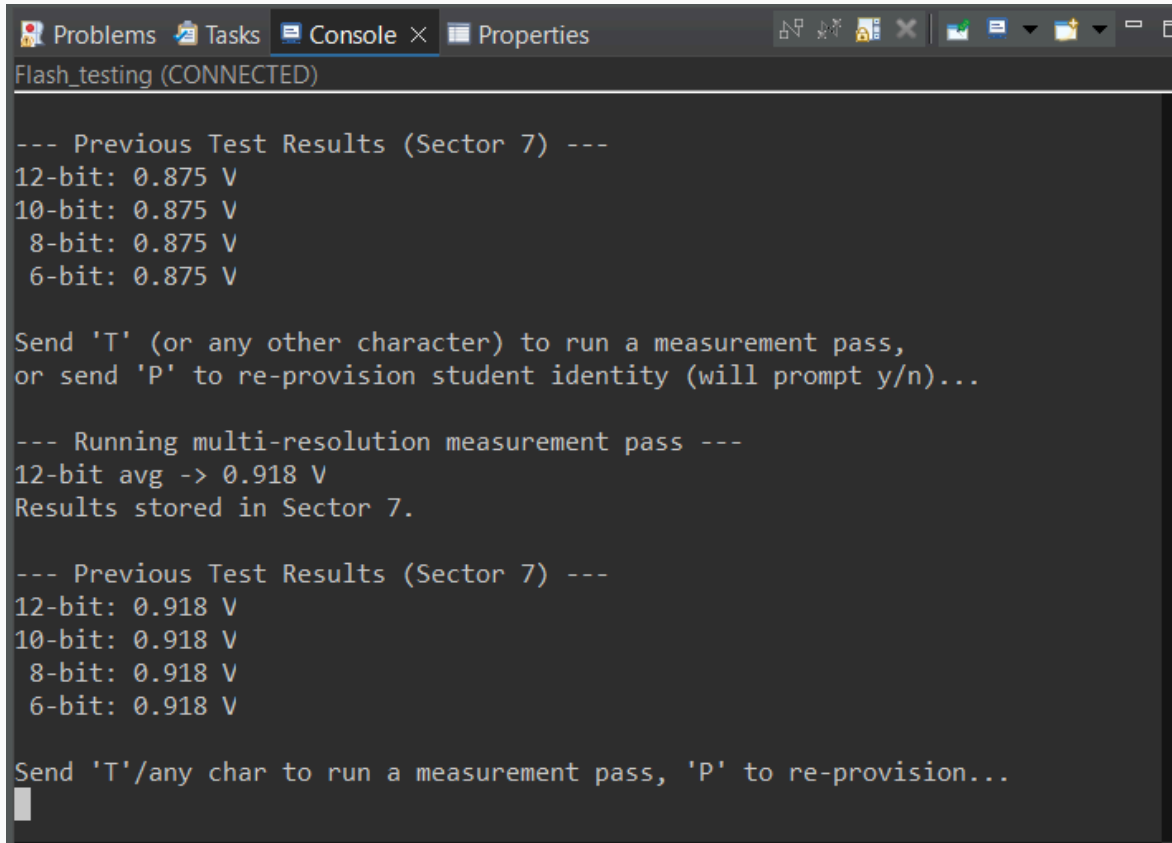
Provisioning is a **one-time** step and never runs on an ordinary boot. In the **register-level build** (LAB_A) it is entered by holding the **B1 USER button (PC13)** down while pressing RESET; in the **HAL build** (LAB_B) it runs automatically only when **Sector 6 has no valid marker and** can be re-run at any time **by sending 'P'**. The routine then:

- Asks for y/n confirmation, warning that Sector 6 will be erased.
- Reads the registration number, roll number, and name over UART, with local echo and backspace editing.
- Builds a 64-byte StudentInfo_t in RAM and stamps its marker with 0xB1010001.
- Erases Sector 6, then programs the record one 32-bit word at a time, and re-locks Flash.

From then on, every boot simply reads the marker at the base of Sector 6 and prints the stored identity.



```
==== HAL Lab 04: ADC Multi-Resolution + Flash Logging ====
-- Student Identity (Sector 6) ---
egistration : 2023715947
oll No. : 04
ame : Syed Muhtasim Apon
-- Previous Test Results (Sector 7) ---
2-bit: 0.875 V
0-bit: 0.875 V
8-bit: 0.875 V
6-bit: 0.875 V
end 'T' (or any other character) to run a measurement pass,
r send 'P' to re-provision student identity (will prompt y/n)...
```



The screenshot shows a terminal window titled 'Flash_testing (CONNECTED)'. It displays the following text:

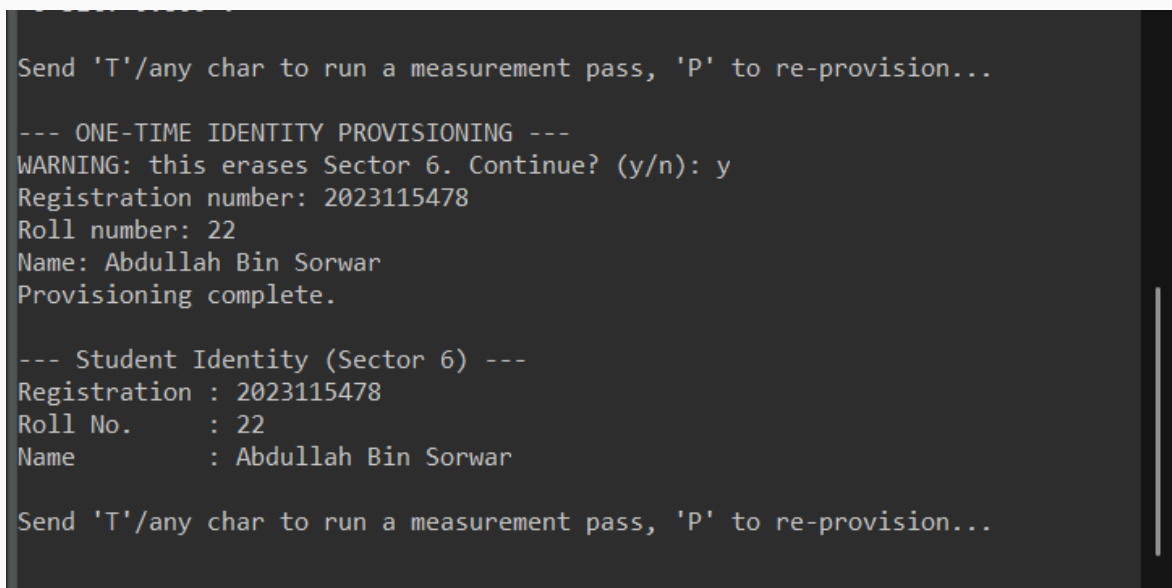
```
--- Previous Test Results (Sector 7) ---
12-bit: 0.875 V
10-bit: 0.875 V
 8-bit: 0.875 V
 6-bit: 0.875 V

Send 'T' (or any other character) to run a measurement pass,
or send 'P' to re-provision student identity (will prompt y/n)...

--- Running multi-resolution measurement pass ---
12-bit avg -> 0.918 V
Results stored in Sector 7.

--- Previous Test Results (Sector 7) ---
12-bit: 0.918 V
10-bit: 0.918 V
 8-bit: 0.918 V
 6-bit: 0.918 V

Send 'T'/any char to run a measurement pass, 'P' to re-provision...
█
```



The screenshot shows a terminal window with the following text:

```
Send 'T'/any char to run a measurement pass, 'P' to re-provision...

--- ONE-TIME IDENTITY PROVISIONING ---
WARNING: this erases Sector 6. Continue? (y/n): y
Registration number: 2023115478
Roll number: 22
Name: Abdullah Bin Sorwar
Provisioning complete.

--- Student Identity (Sector 6) ---
Registration : 2023115478
Roll No.      : 22
Name          : Abdullah Bin Sorwar

Send 'T'/any char to run a measurement pass, 'P' to re-provision...
```

Figure — Terminal screenshot of a provisioning session (entering registration / roll / name).

5. Boot Output (Sample)

```
===== Lab 04: ADC Multi-Resolution + Flash Logging =====

--- Student Identity (Sector 6) ---
Registration : 2023715947
Roll No.      : 04
```

Name : Syed Muhtasim Apon

--- Previous Test Results (Sector 7) ---

12-bit: 1.620 V

10-bit: 1.631 V

8-bit: 1.600 V

6-bit: 1.571 V

```
===== HAL Lab 04: ADC Multi-Resolution + Flash Logging =====
```

```
--- Student Identity (Sector 6) ---
```

```
Registration : 2023715947
```

```
Roll No. : 04
```

```
Name : Syed Muhtasim Apon
```

```
--- Previous Test Results (Sector 7) ---
```

```
12-bit: 0.793 V
```

```
10-bit: 0.793 V
```

```
8-bit: 0.793 V
```

```
6-bit: 0.793 V
```

```
Send 'T' (or any other character) to run a measurement pass,  
or send 'P' to re-provision student identity (will prompt y/n)...
```

Figure — Terminal screenshot of the boot sequence on your board.

7. Test Cases (TC1–TC12)

N.B.:

Only the flash memory part has been executed. That's why only test cases related to flash memory have been conducted and observed. So test cases related to ADC na DAC are not applicable

#	Test case	Steps	Expected result	Observed / actual	Verdict
TC 1	Raw ADC responsiveness	N/A	N/A	N/A	N/A
TC 2	Flash erase verification	Erase Sector 7, read base address	0xFFFFFFFF	Base read back 0xFFFFFFFF	Pass
TC 3	Flash write / read-back	Write 0xDEADBEEF to base, read back	Reads exactly 0xDEADBEEF	Read back 0xDEADBEEF	Pass

#	Test case	Steps	Expected result	Observed / actual	Verdict
TC 4	Identity provisioning	Run the one-time provisioning routine	Marker + reg + roll + name written; read-back matches input	Details entered; read-back matched exactly	Pass
TC 5	Identity display at boot	Reset board after provisioning	Reg / roll / name displayed first, before anything else	Identity printed first on reset	Pass
TC 6	Identity survives testing	N/A	N/A	N/A	N/A
TC 7	Blank results fallback	Erase Sector 7 only, reboot without a test	Shows "no previous test data"; no crash	N/A	N/A
TC 8	Per-resolution averaging	Log raw samples per resolution vs stored average	Stored average matches independent average of logged samples	N/A	N/A
TC 9	Result overwrite on re-test	Run suite twice via two triggers, POT moved between	2nd run's four voltages shown after next reset; no old/new mixing	N/A	N/A
TC 10	Resolution boundary sanity	Inspect stored voltages at all four resolutions, same POT	All four reasonably close (within quantisation error)	N/A	N/A
TC 11	UART auto-trigger reliability	Single keypress, then a rapid burst, separately	Each starts exactly one full 4-res run; no miss / duplicate	N/A	N/A
TC 12	Power-cycle persistence	Unplug/replug USB after provisioning + testing	Identity and last results both survive and display	Both survived unplug/replug; displayed OK	Pass

```

===== HAL Lab 04: ADC Multi-Resolution + Flash Logging =====

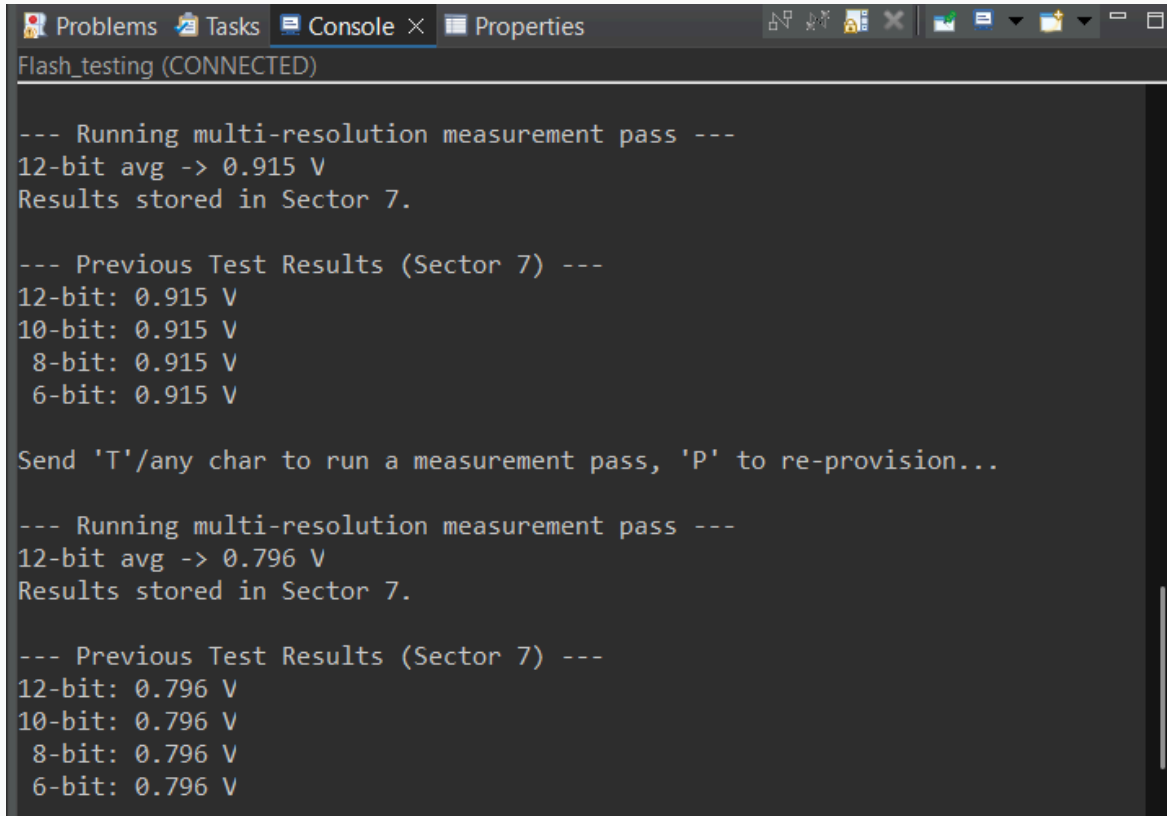
--- Student Identity (Sector 6) ---
Registration : 2023715947
Roll No.    : 04
Name       : Syed Muhtasim Apon

```

```
Send 'T' (or any other character) to run a measurement pass,  
or send 'P' to re-provision student identity (will prompt y/n)...  
  
--- ONE-TIME IDENTITY PROVISIONING ---  
WARNING: this erases Sector 6. Continue? (y/n): Y  
Registration number: 2023115948  
Roll number: 22  
Name: Abdullah Bin Sorwar  
Provisioning complete.  
  
--- Student Identity (Sector 6) ---  
Registration : 2023115948  
Roll No.      : 22  
Name          : Abdullah Bin Sorwar  
  
Send 'T'/any char to run a measurement pass, 'P' to re-provision...  
█
```

Figure 7.1 — Evidence screenshots for TC1–TC6 (raw sweep, Flash erase/write, provisioning, boot display).

```
Flash_testing (CONNECTED)  
6-bit: 0.793 V  
  
Send 'T' (or any other character) to run a measurement pass,  
or send 'P' to re-provision student identity (will prompt y/n)...  
  
===== HAL Lab 04: ADC Multi-Resolution + Flash Logging =====  
  
--- Student Identity (Sector 6) ---  
Registration : 2023115948  
Roll No.      : 22  
Name          : Abdullah Bin Sorwar  
  
--- Previous Test Results (Sector 7) ---  
12-bit: 0.793 V  
10-bit: 0.793 V  
8-bit: 0.793 V  
6-bit: 0.793 V  
  
Send 'T' (or any other character) to run a measurement pass,  
or send 'P' to re-provision student identity (will prompt y/n)...
```



```
Problems Tasks Console × Properties
Flash_testing (CONNECTED)

--- Running multi-resolution measurement pass ---
12-bit avg -> 0.915 V
Results stored in Sector 7.

--- Previous Test Results (Sector 7) ---
12-bit: 0.915 V
10-bit: 0.915 V
8-bit: 0.915 V
6-bit: 0.915 V

Send 'T'/any char to run a measurement pass, 'P' to re-provision...

--- Running multi-resolution measurement pass ---
12-bit avg -> 0.796 V
Results stored in Sector 7.

--- Previous Test Results (Sector 7) ---
12-bit: 0.796 V
10-bit: 0.796 V
8-bit: 0.796 V
6-bit: 0.796 V
```

Figure 7.2 — Evidence screenshots for TC7–TC12 (blank fallback, averaging, overwrite, persistence).

8. Analysis Questions

Q1. Why does Flash require an erase-before-write cycle, while RAM does not?

- A flash bit cell is a **floating-gate transistor**: It works like an **electrically isolated gate**. It injects charge to change a 1 into a 0. But cannot charge back a 0 into a 1. Restoring bits to 1 requires an **erase**, which removes the whole sector at once (all bits → 1).
- SRAM, by contrast, is a **bistable latch** (cross-coupled transistors) with no charge trapping; **each cell can be driven to 0 or 1 freely and independently on every write**, so no erase step is needed.

Q2. Sector 7 endurance (≈10,000 cycles) — how long would it last, and how to reduce wear?

- Every test run erases Sector 7 once (one erase cycle). If a run were triggered every ~5 seconds continuously, that is 12 erases/minute ≈ **720 erases/hour**. The 10,000-cycle budget would then

be exhausted in about $10,000 \div 720 \approx 14$ hours of continuous use — well under a single day. (Even at one run per minute it lasts only ~7 days.)

- Fix — **log-structured / append-based wear-levelling**: instead of erasing and rewriting the same word every run, append each new `ResultsRecord_t` at the next free offset within the sector and erase only when the sector is full. A 128 KB sector holds thousands of small records, so one erase now covers thousands of writes — multiplying endurance by roughly that factor.

Q3. Why two separate sectors instead of one? What would go wrong if shared?

- Erase is **whole-sector**, and the two data kinds have opposite lifecycles: identity is written once, while results are rewritten on every run. If both lived in the same sector, **each results update would have to erase that sector — which would also wipe the identity block**.
- Then be forced to rewrite the identity on every run (**extra wear and complexity**), the “write-once” guarantee would be lost, and a power failure during the erase could destroy the identity permanently. Separating them means erasing Sector 7 can never touch Sector 6, so testing is always safe (TC6).

Q4. Risk of losing power mid erase/write, and how a robust design protects the results block.

- If power is lost part-way through erasing or programming Sector 7, the block is left in an **inconsistent state**: the sector may be erased but not yet reprogrammed (**reads as `0xFFFFFFFF` → looks blank**), or a record may be half-written (a valid-looking marker over garbage data). The stored results can be **lost or silently corrupt**.
- A more robust design (conceptually): **to write the payload first and commit the validity marker last**. Then combining this with **ping-pong (A/B) slots or append-with-checksum records**: each record carries a CRC and a sequence number, and on boot the firmware picks the newest record whose CRC validates — a corrupted in-progress record is ignored and the previous good record survives. A brown-out detector (PVD) can additionally refuse to start a Flash write when VDD is sagging